

B.Tech Industrial Training Presentation on AIR INDIA INTERNSHIP



Presented by

HARSH AGARWAL(201600622)

MONODEEP SAHA(201600523)

Group No: 24

E&C Engg. Dept., SMIT

Under the guidance of

A.CHAKRABARTI

A.G.M(SS)- Training, AIESL, AIR INDIA

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INTRODUCTION

An aircraft is an engineering marvel. Its spectrum of use ranges from transportation, recreation, research to military applications. It is an assimilation of multiple instruments which are varying in operation. In this presentation, we look at the electrical, electronic and radio communication wings of a modern commercial aircraft under operation with Air India Limited.

INSTRUMENT OVERHAUL

Course Director Indicator(CDI)

- It displays a pictorial plane view of an aircraft with respect to magnetic north, to the selected course and selected heading.

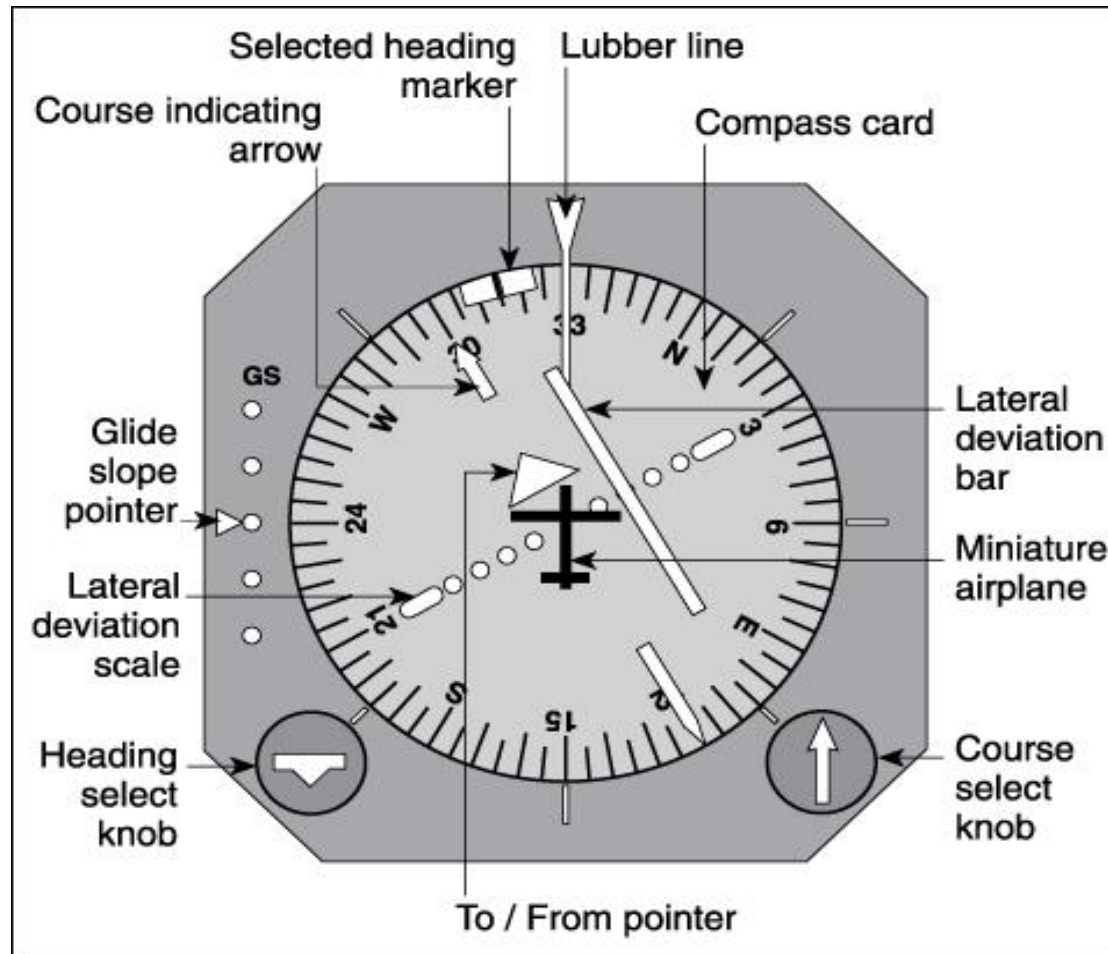
Flight Director Indicator(FDI)

- It tells the pilot of an aircraft, the attitude required to follow a certain trajectory to which the flight is to be conducted.

COCKPIT INSTRUMENTS SYSTEM



COURSE DIRECTOR INDICATOR



FLIGHT DIRECTOR INDICATOR

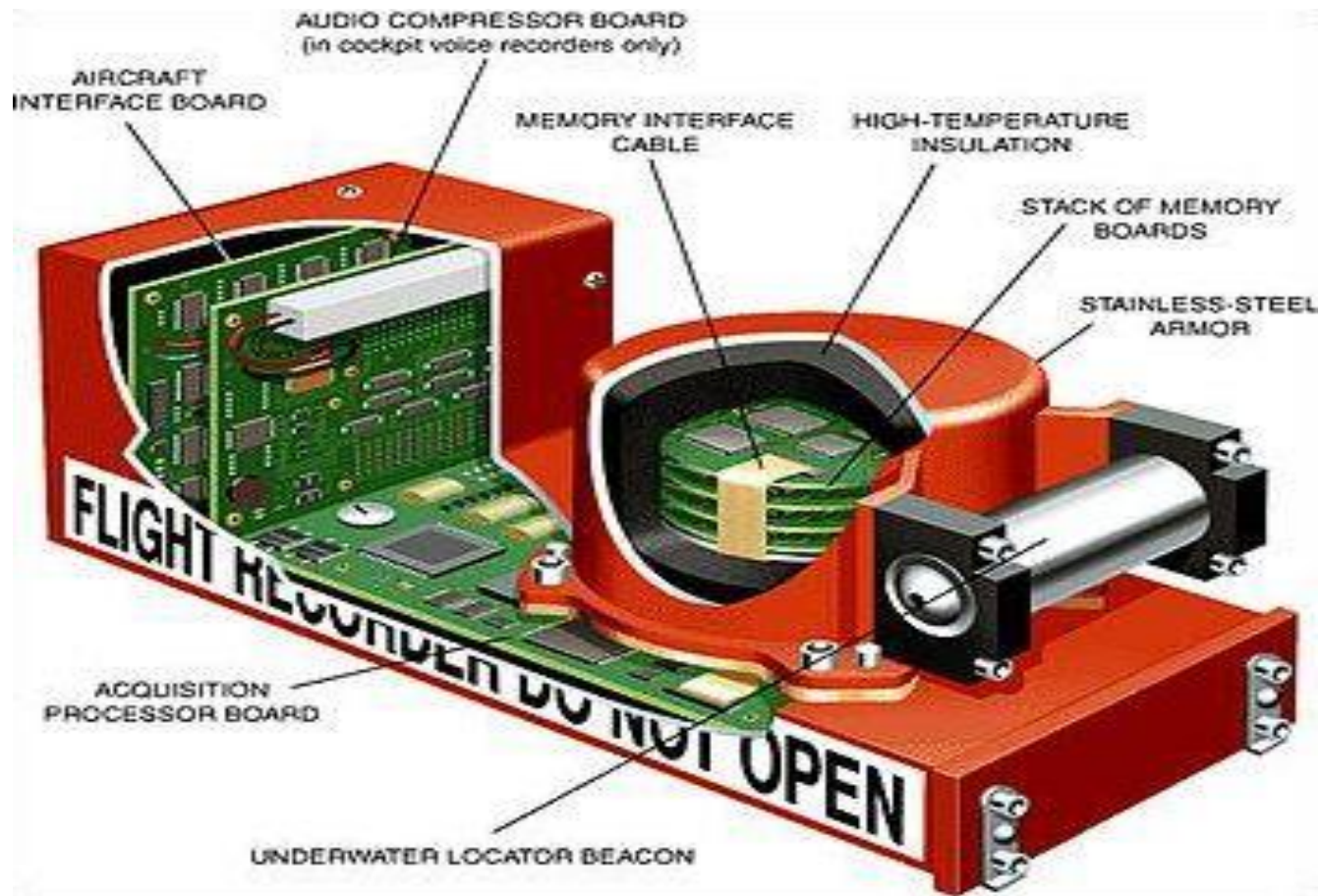


RADIO OVERHAUL

Cockpit Voice Recorder(CVR)

- A cockpit voice recorder (CVR) is a flight recorder used to record the audio environment in the flight deck of an aircraft for the purpose of investigation of accidents and incidents.
- It is popularly termed as black box. It has a memory unit to process the audio inputs and memory units to store them.

Solid State Cockpit Voice Recorder



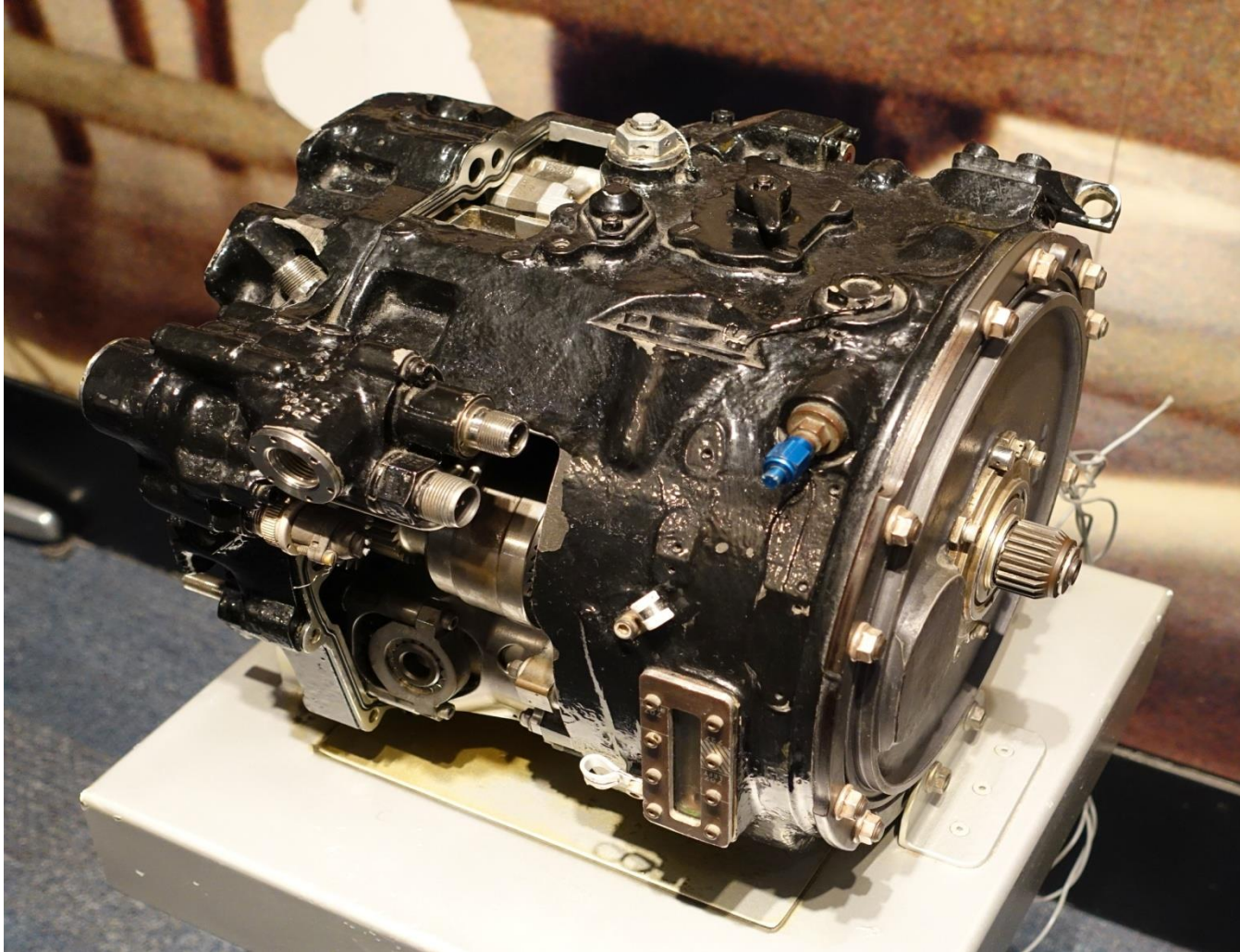
EMERGENCY LOCATION TRANSMITTER

The ELT is a self contained transmitter that is designed to help locate an airplane after crash. The ELT is automatically turned on by crash forces. It has a floatation device and a flexible antenna. It is of two types: Crash activated and Water activated. It emits a digital distress signal to satellites which is used for accurate localization and identification.

ELECTRIC OVERHAUL

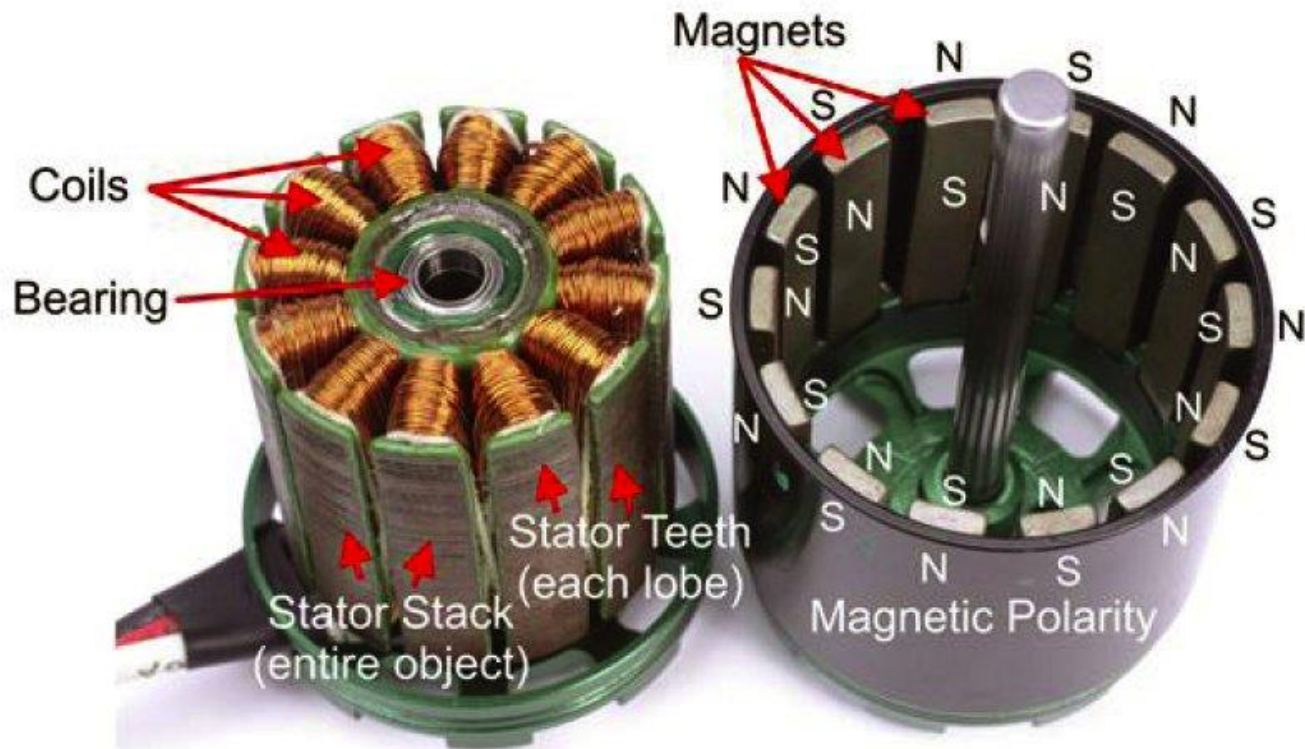
An aircraft electrical system is a self contained network of components that generate, transmit, distribute, utilize and store electrical energy. All aircraft electrical systems have components with the ability to generate electricity. Depending upon the aircraft, generators or alternators are used to produce electricity. These are usually engine driven but may also be powered by an APU, a hydraulic motor or a Ram Air Turbine (RAT).

CONSTANT SPEED DRIVE



DC MOTOR

OUTRUNNER COMPONENTS



Conclusion And Future Scope

Aeroplane is among the greatest inventions of the human mind. Its sheer enormity and the diversity of its components make the study of an aircraft and its operating principles a long and difficult journey. The time constraints under which this industrial training was pursued made it even more challenging. Nonetheless, we hope that however much could have been studied in this time; we have been successful in doing so. Being able to get a basic knowledge of the electrical, electronics, radio, communication, mechanical, instrumental aspects of an aircraft has been a highly satisfying and enriching experience.

ACKNOWLEDGEMENT

I would like to take this opportunity to express my gratitude towards Air India Limited for providing me with this wonderful opportunity of interning with it. I would also like to thank all the respective and respected teachers of the various departments who shared their invaluable wisdom with me and my group. I am especially grateful towards the different shop in-charges who were always there to guide us during our internship and helped us in getting to know the subject matter more proficiently. Finally, I am appreciative of my family and friends for helping me out throughout this industrial training.

References (Following IEEE Chicago format)

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Thank You